

Improved Grid Forming Control for Energy-Type STATCOM to Enhance Damping and Limit Multiple Currents

Xianqiang Shi [✉], *Member, IEEE*, Rui Li, *Senior Member, IEEE*, Xiqi Wu [✉], *Student Member, IEEE*, Dangsheng Zhou, *Member, IEEE*, Yirun Cheng, and Xu Cai [✉], *Senior Member, IEEE*

Abstract—To improve the frequency and voltage stability of the modern power system, an energy-type STATCOM (ET-STATCOM) formed by embedding supercapacitors (SCs) in the dc bus of two film capacitor-type STATCOMs was designed, which can provide virtual inertia by boosting active power and voltage support through reactive power compensation. However, the issue of its grid forming control hangs in the balance. First, the favored virtual synchronous machine control may lead to over-charge/-discharge of SCs due to the active power-based control target. Second, although the classical matching control (MC) can regulate SC energy, none of its existing damping enhancement schemes is proper for the ET-STATCOM. Last but not least, the existing current limiting strategies can only limit the total current magnitude synthesized by the active and reactive currents, but not their respective magnitudes. To address these issues, an energy synchronization link founded on the MC and a novel damping enhancement scheme based on active power direct feedback are proposed, which helps the ET-STATCOM to integrate self-synchronization with power oscillation suppression capability, spontaneous inertia response, and autonomous management of SC energy. In addition, a multicurrent limiting control technique has been investigated to achieve overall and independent limiting of active and reactive currents. Simulations and a practical 35kV/30MVar/3MW/4s prototype validate the method's effectiveness.

Index Terms—Current limiting, damping enhancement scheme, energy-type STATCOM (ET-STATCOM), grid-forming control (GFMC), matching control (MC), SC energy management, virtual synchronous machine (VSM).

I. INTRODUCTION

NOWADAYS, power converters have become essential in modern power systems due to the widespread integration of renewable energy sources [1], [2]. However, their increasing replacement of synchronous generators (SGs) raises stability concerns, as power converters lack kinetic energy [3]. This significantly reduces system inertia, threatening grid dynamic

stability [4]. Consequently, ensuring adequate inertia and reactive power support to maintain frequency and voltage within allowable limits has become a critical challenge [5].

Configuring energy storage systems (ESSs) with proper power control to emulate inertia has proven effective in power system applications [6]. The utilization of film capacitor-type STATCOM (FC-STATCOM) in renewable power plants to simulate inertia is proposed in [7] and [8]. However, on the one hand, the energy stored inside the FC-STATCOMs is inherently limited. On the other hand, dc link voltage variations should be limited to avoid over-modulation [9], implying that the energy stored in FCs cannot be fully released. In other words, FC-STATCOMs exhibit faint inertial response capability. Battery ESS (BESS) has emerged as an attractive candidate for power buffers and peak load shifting due to its high energy density characteristics [10]. Theoretically, the BESS could also be used downstream as a medium for inertial emulation. However, the frequent charging and discharging events may adversely impact its lifespan [11]. Compared with FCs and batteries, supercapacitors (SCs) feature high power density, long lifespan, and fast charging speed. Hence, SC-based ESS is a promising candidate for inertia emulation [12].

To construct a high-capacity, high-reliability SC-based ESS, a straightforward approach is to adopt the classical cascaded H-bridge architecture by integrating SCs into the cascaded sub-modules [13]. While simple to implement, this method suffers from low SC energy utilization, similar to FC-STATCOMs. To overcome this limitation, a novel solution proposes embedding SCs into the dc bus of two FC-STATCOMs, forming an energy-type STATCOM (ET-STATCOM), as illustrated in Fig. 1. Industry leaders such as Siemens, ABB, Sieyuan Electric Co., Ltd., and NR Electric Co., Ltd. have branded this technology as “SVC PLUS FS (E-STATCOM)” [14], “SVC Light Enhanced” [15], and “static synchronous condenser” [16], [17], respectively.

It is now widely recognized that applying grid-forming control (GFMC) to power electronic converters can enhance weak grid stability while providing natural frequency and voltage regulation services [18], [19]. Regarding its synchronization link, i.e., active power-frequency control link, mainstream GFMC strategies can be categorized into two types based on their control objectives, i.e., active power control type and dc voltage control type. The former controls the active power of the

Received 12 December 2024; revised 14 February 2025 and 23 April 2025; accepted 11 May 2025. Date of publication 13 May 2025; date of current version 5 August 2025. Recommended for publication by Associate Editor S. Mekhilef. (Corresponding author: Xu Cai.)

Xianqiang Shi, Rui Li, Xiqi Wu, Yirun Cheng, and Xu Cai are with the School of Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: xianqiang.shi@sjtu.edu.cn; liruiqd@sjtu.edu.cn; wux-xiqi@sjtu.edu.cn; chengyirun1028@sjtu.edu.cn; xucai@sjtu.edu.cn).

Dangsheng Zhou is with the Shenzhen Hopewind Electric Company, Ltd., Shenzhen 518055, China (e-mail: zhou dangsheng@hopewind.com).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPEL.2025.3569931>.

Digital Object Identifier 10.1109/TPEL.2025.3569931

0885-8993 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See <https://www.ieee.org/publications/rights/index.html> for more information.

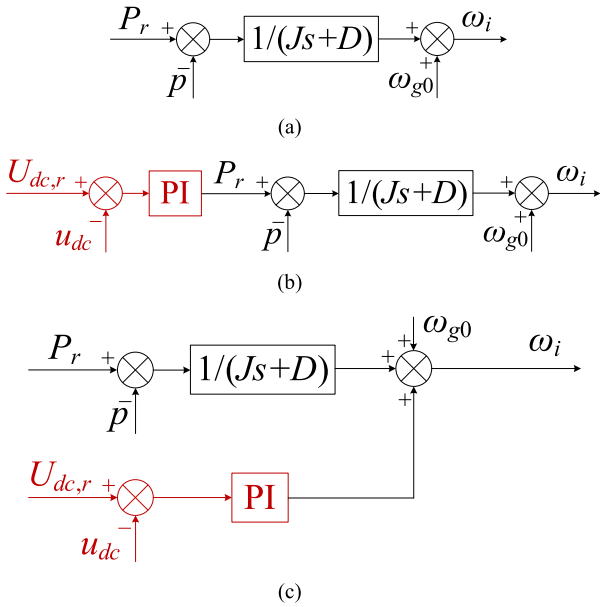


Fig. 1. Control block diagram of the classical VSM and two modified VSMs. (a) Classical VSM. (b) Cascade-type modified method. (c) Parallel-type modified method.

converter under the assumption that the dc voltage is constant or well controlled [20]. A typical example of this control strategy is the conventional virtual synchronous machine (VSM) [21], [22], with its control block diagram illustrated in Fig. 1(a). The VSM has been widely adopted as a grid interface for the BESS. Ostensibly, it can be applied to the ET-STATCOM as well. However, it is known that the SC voltage increases when charging and decreases when discharging. As a result, the ET-STATCOM will suffer from two fatal flaws in its adoption of the VSM. On the one hand, its damping term is coupled to the primary frequency regulation, which should be avoided due to the limited energy stored in SCs. On the other hand, the SC voltage in this control strategy is left uncontrolled, which may result in over-charge/over-discharge of SCs. To address the same problems with FC-STATCOM's application of the methodology, researchers have suggested cascading a dc voltage regulation loop with the classical VSM to generate its active power reference [23], as illustrated in Fig. 1(b) or, alternatively, connecting the dc voltage regulation loop in parallel with the VSM to jointly determine the converter output frequency [24], as shown in Fig. 1(c). Unfortunately, these methods are inappropriate for the studied ET-STATCOM.

Apparently, both cascade-type and parallel-type modified VSMs require at least four parameters (J , D , and two PI controller gains), increasing system complexity and parameter tuning difficulty. Fortunately, the primary objective of implementing GFMC in FC-STATCOMs is to enhance weak grid stability by eliminating PLL dependence, rather than providing inertial support, given its limited energy storage capacity. Thus, the methods in Fig. 1(b) and (c) are appropriate for FC-STATCOMs as they can maintain stable FC voltages for modulation requirements while simplifying parameter tuning by focusing mainly on stability [25], [26]. In contrast, ET-STATCOMs are designed

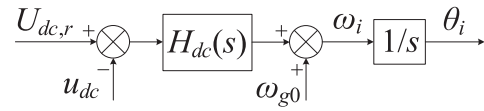


Fig. 2. Control block diagram of the MC's DC voltage control loop.

to inject substantial active power for grid frequency support, not to maintain constant SC voltage for modulation. When applying these methods, ET-STATCOMs must balance four parameters across multiple objectives: system stability, response dynamics, inertial support capability, and SC energy distribution. Consequently, the approaches in Fig. 1(b) and (c) are less friendly for ET-STATCOMs.

To address the above issues, the dc voltage control type GFMC is studied, which utilizes the converter's output frequency to match dc capacitor dynamics with SG dynamics for PLL-less synchronization. The renowned matching control (MC) technique is a leading example of this control approach [27]. It should be noted that the complete MC contains two parts, i.e., the dc voltage control loop shown in Fig. 2 and its corresponding damping enhancement loop. A comparison of Figs. 1(b), (c), and 2 reveals that the MC technique is significantly simplified, thereby improving the dynamic performance of dc voltage regulation and grid synchronization [9], [25]. In Fig. 2, $H_{dc}(s)$ denotes the dc voltage controller, which can be designed as a proportional coefficient [25], a lead-lag compensator [28], or a PI controller [29]. The key distinction lies in whether the dc voltage can revert to its setpoint despite grid frequency deviations. Only the last scenario achieves this, whereas the first two fail. The MC technique is mainly oriented towards back-to-back converters, where the grid-side converter (GSC) uses it to control the dc bus voltage, while the active power is left for the machine-side converter (MSC). However, while the method performs well for the back-to-back converters, further research is essential to evaluate its applicability and implementation in single-ended converters like the studied ET-STATCOM.

As indicated above, the MC shown in Fig. 2 is undamped, which may lead to system oscillations and instability. To solve this problem, scholars have proposed several damping enhancement methods, which are realized in three different channels, namely:

- 1) the active power channel of the MSC [27],
- 2) the voltage output channel of the GSC [25], [29], and
- 3) the phase angle output channel of the GSC [28], as shown in Fig. 3, respectively.

As depicted in Fig. 3(a), this approach enhances the MC damping by modifying the active power reference of the MSC. Although this method is highly effective, it is inapplicable to the studied ET-STATCOM as it is a single-ended converter lacking an MSC. The second damping enhancement method, illustrated in Fig. 3(b), is inspired by the power system stabilizer (PSS) concept in SGs. The PSS measures the SG's angular speed, processing it through an amplifier and a high-pass filter, and injects the output as a supplementary signal into the excitation system. Adopting a similar approach, this method enhances damping in the MC by feeding the dc bus voltage deviation into

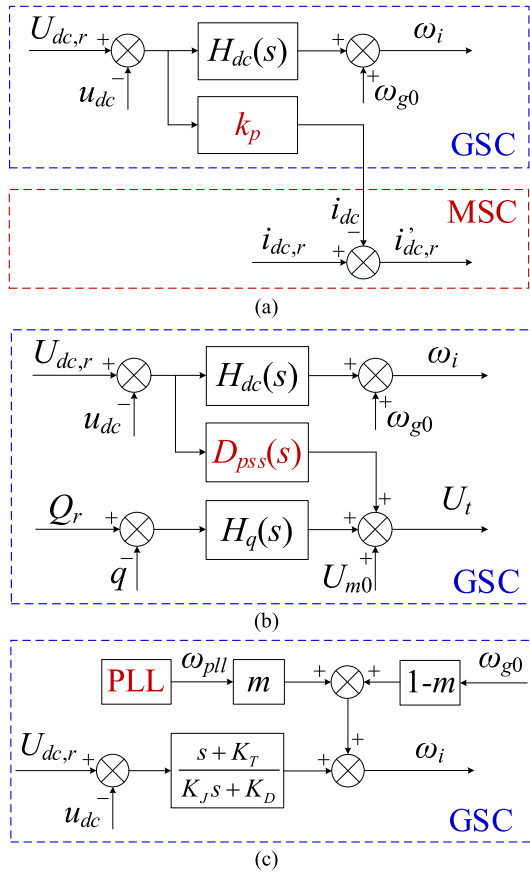


Fig. 3. Methods for damping enhancement in the MC's DC voltage control loop. (a) Active power adjustment of the MSC. (b) Output voltage adjustment of the GSC. (c) Phase angle adjustment of the GSC's output.

the converter output voltage. The resulting voltage amplitude will generate damping power, effectively emulating the damping mechanism of an SG. However, the enhanced damping magnitude is influenced by the steady-state power angle operating point. When no active power is output (zero power angle), this additional damping term will vanish [30]. Coincidentally, the ET-STATCOM is only capable of inertial response, at the end of which it no longer outputs active power to the outside, so this damping compensation strategy will fail in the new steady state. As a result, the method is not friendly to the ET-STATCOM either. It can be seen from Fig. 3(c) that a lead-lag compensator is cascaded in the phase angle output channel to improve the damping characteristics of the MC. However, the permissible dc voltage deviation corresponding to grid frequency variations is coupled with the active power output capability (i.e., inertial response capability) through two critical parameters, K_T and K_D , as depicted in Fig. 3(c). To achieve decoupling, the frequency generated by the PLL is added again to the converter output frequency with a tradeoff. Obviously, the method fails to get rid of PLL completely, so the stability of its weak grid operation is questionable.

In summary, none of the existing GFMC frameworks are appropriate to the ET-STATCOM. Therefore, it is necessary to investigate an improved GFMC scheme for the ET-STATCOM.

In addition to the synchronization link analyzed above, the current limiting control is also indispensable for the GFMC framework. Literature [31] categorizes the common current limiting methods in the GFMC framework into three groups, i.e., direct current limiting, indirect current limiting, and hybrid current limiting. Although the above-mentioned techniques work well for wind turbine inverters, BESSs, etc., they are inadequate for ET-STATCOM applications. This is because all the aforementioned current-limiting strategies can only constrain the total current amplitude (i.e., the synthesized vector of active and reactive currents), but not their individual magnitudes. The active and reactive currents in the ET-STATCOM are supplied by the SCs on the MMC's dc bus and by the FCs embedded inside the MMC, respectively (see Fig. 1) so that we can configure and regulate its active and reactive powers independently, which is one of the main advantages of the topology. Due to the high cost of SCs and challenges in series-parallel boosting and capacity expansion [15], [17], the active capacity of ET-STATCOM is typically much lower than its reactive capacity. Consequently, merely limiting the total current amplitude fails to ensure simultaneous protection for both SCs and MMC. Despite its importance, this issue remains understudied.

Motivated by the gap in research, this article explores an improved GFMC for the ET-STATCOM to enhance its damping characteristics and limit multiple currents. The main contributions are summarized as follows:

- 1) An energy synchronization link based on the MC concept is proposed for the ET-STATCOM, leveraging SC energy dynamics to achieve self-synchronization and autonomous management of SC energy. The study thoroughly investigates the implementation, inertia emulation mechanism, SC energy distribution characteristics, and design methodology for key control parameters.
- 2) A novel damping enhancement method for the energy synchronization link based on active power direct feedback (APDF) is proposed. The method is independent of the active power level of the equipment and does not involve PLL. Moreover, it does not alter the self-synchronization characteristics, inertial response characteristics, and SC energy distribution characteristics of the energy synchronization link. The proposed complete synchronization methodology enables the organic combination of PLL-less self-synchronization with power oscillation suppression capability, spontaneous inertia response, and autonomous management of SC energy.
- 3) Aiming at the drawback that the existing current limiting strategies can only limit the total current amplitude, a multiple currents limiting control (MCLC) technique is investigated to achieve a comprehensive limiting capability of the total current magnitude as well as active and reactive currents to ensure the safety of the MMC and SCs.

The rest of this article is as follows. Section II describes the architecture, operating principle, and dynamical models of the ET-STATCOM. Section III presents the proposed improved GFMC, including the MC-based energy synchronization link, the APDF-based damping enhancement scheme, and

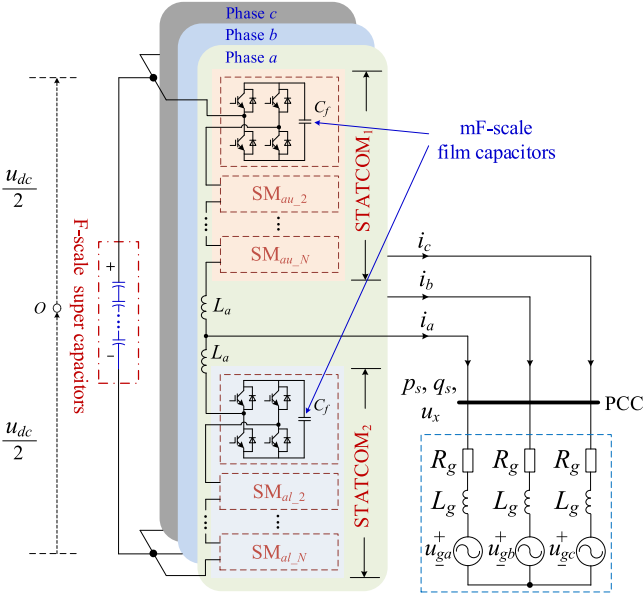


Fig. 4. Main circuit configuration of the studied ET-STATCOM.

the MCLC method. Then, the small signal modeling and control design is studied in Section IV. Comparative simulations based on the PSCAD/EMTDC platform and experiments from a 35 kV/30 MVar/3WM/4s prototype are provided in Sections V and VI, respectively. Finally, Section VII concludes this article.

II. BASIC THEORY OF THE ET-STATCOM

Fig. 4 illustrates the structure of the studied ET-STATCOM, which consists of six arms, each containing N full-bridge sub-modules (FBSMs). Each FBSM comprises four IGBTs/diodes and a millifarad (mF)-scale FC. The farad (F)-scale SCs are connected to the dc bus of the two FC-STATCOMs. In fact, the two FC-STATCOMs form an FBSM-type modular multilevel converter (MMC). The ET-STATCOM can operate in a four-quadrant mode, enabling simultaneous frequency stabilization and voltage regulation.

This topology offers the following advantages.

- 1) This design ensures stable MMC modulation performance regardless of SC voltage fluctuations. Even if the SC voltage drops to zero, the MMC maintains normal operation, enabling a wide operating voltage range for the SCs and enhancing inertial response capability.
- 2) The active and reactive power flows are decoupled in this system. When supplying reactive power, the SCs operate equivalently as disconnected from the grid, reducing their cycling frequency and extending their lifespan. In addition, the system allows independent control of active and reactive power magnitudes for enhanced flexibility. Typically, the active power capacity is lower than the reactive capacity due to technological limitations in SC series-parallel voltage boosting and capacity expansion, as well as cost considerations.

Fig. 5 illustrates the equivalent circuit of the ET-STATCOM, depicting the active power transfer path from the dc bus to

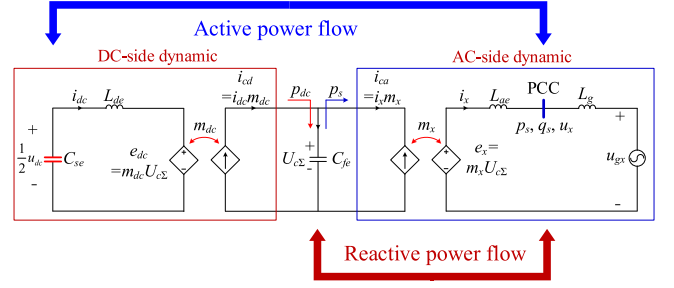


Fig. 5. Equivalent circuit of the studied ET-STATCOM.

the ac grid via the MMC. This representation follows directly from the fundamental operating principles of MMCs [32], [33]. Notably, this article primarily focuses on investigating the external grid-connection characteristics of the ET-STATCOM. For detailed discussions on internal MMC control aspects, such as phase and arm capacitor voltage balancing, circulating current suppression, and modulation techniques, please refer to [32] and [33] for more details.

In Fig. 5, u_{dc} and i_{dc} represent the dc bus voltage and current, respectively; L_{de} and L_{ae} denote the dc-side and ac-side equivalent inductance, respectively; e_{dc} is the dc component of the modulating signal and e_x is the ac component of phase x ($x = a, b, c$) of the modulating signal; m_{dc} and m_x denote the dc component and ac component of the modulated signal of phase x ($x = a, b, c$), respectively; i_{cd} and i_{ca} represent the dc component and ac component of the current flowing through the FCs, and they are equal to $i_{dc}m_{dc}$ and i_xm_x , respectively; $U_{c\Sigma}$ indicates the aggregated voltage of the FCs, and C_{fe} denotes its aggregated capacitance; p_{dc} and p_s are the dc-side and ac-side active power, respectively; u_x and u_{gx} denote the voltages across the point of common coupling (PCC) and the power grid, respectively; i_x is the current of phase x flowing into the power grid; C_{se} denotes the aggregated capacitance of the SCs.

Obviously, the current and voltage of the SCs satisfy the following dynamic equations:

$$C_{se} \frac{du_{dc}}{dt} = -i_{dc}. \quad (1)$$

When the total energy of the SM capacitor is controlled to be constant [32], the relationship between the SC energy and the active power p_s can be expressed as

$$\frac{dw_{ts}}{dt} = -p_{dc} = -p_s = -u_{dc}i_{dc} \quad (2)$$

where w_{ts} denotes the actual value of the total energy stored in the SCs, and it satisfies $w_{ts} = C_{se}U_{dc0}^2/2$.

The following variables can be defined

$$T_{sc} = \frac{2W_{ts0}}{P_0} = \frac{C_{se}U_{dc0}^2}{P_0} \quad (3)$$

where W_{ts0} denotes the rated value of the SC energy, and it satisfies $w_{ts} = C_{se}U_{dc0}^2/2$ (U_{dc0} is the rated value of SC voltage or dc bus voltage); T_{sc} is defined as the inertia time constant of SCs; P_0 is the rated active power of the ET-STATCOM.

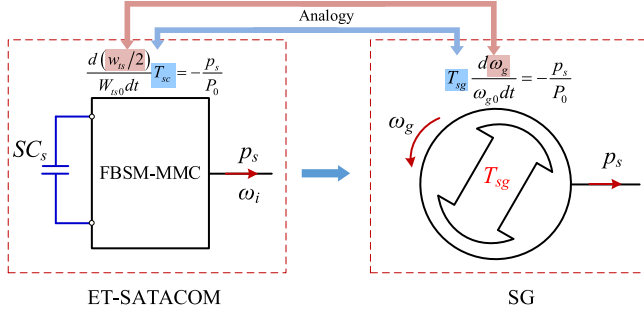


Fig. 6. Analogy relationship between ET-STATCOM and SG.

Substituting (3) into (2), yields

$$T_{sc} \frac{d(w_{ts}/2)}{W_{ts0} dt} = -\frac{p_s}{P_0}. \quad (4)$$

Obviously, the active power of the ET-STATCOM can be mapped in the variation of SC energy, which is similar to the rotor equation of the governor-less SG, i.e.,

$$T_{sg} \frac{d\omega_g}{\omega_{g0} dt} = -\frac{p_s}{P_0} \quad (5)$$

where ω_g and ω_{g0} represent the actual value and rated value of the SG angular frequency, respectively; T_{sg} is the inertia time constant of the SG, and it satisfies $T_{sg} = J_{sg0}\omega_{g0}^2/P_0$ [22], [25] (J_{sg0} is the SG rotor's moment of inertia).

III. IMPROVED GRID FORMING CONTROL FOR ET-STATCOM

A. MC-Based Energy Synchronization Link

The VSM primarily regulates active power, leaving the SC voltage uncontrolled, which can lead to overcharging/discharging. To address this limitation, ET-STATCOM should incorporate MC technology for grid self-synchronization and autonomous SC energy management.

A comparison of (4) and (5) reveals that the SC energy has dynamics similar to the SG frequency. According to the similarity principle of the dynamical system, one-half of the SC energy can be analogous to the converter frequency, and the inertia time constant T_{sc} of SCs in (3) can be analogous to the inertia time constant T_{sg} of the SG. The relationships between these preceding variables are shown in Fig. 6.

Unlike the PLL-based control structure, the active power-frequency and reactive power-voltage amplitude control are used in the GFMC framework [19], [20], [21], [22]. Considering an inductive power grid, the relationships of active/reactive power, power angle, and voltage amplitude can be established as

$$\begin{cases} p_s = \frac{3U_m U_{gm}}{2X_g} \sin \delta \\ q_s = \frac{3U_m (U_m - U_{gm} \cos \delta)}{2X_g} \end{cases} \quad (6)$$

where δ denotes the power angle between the PCC voltage and the grid voltage, satisfying the relation $s\delta = \omega_i - \omega_g$; U_{gm} and U_m represent the amplitudes of the PCC voltage and grid voltage, respectively; X_g is the reactance between the PCC and the power grid, given by $X_g = \omega_g L_g$.

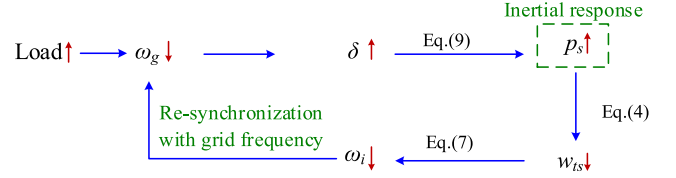


Fig. 7. Schematic diagram of grid self-synchronization and spontaneous inertial response of the MC-based energy synchronization link.

According to Fig. 6, a relationship between the ET-STATCOM output frequency and the SC energy can be established as

$$\frac{w_{ts} - W_{ts0}}{2W_{ts0}} = k_s \frac{\omega_i - \omega_{g0}}{\omega_{g0}} \quad (7)$$

where ω_i is the output angular frequency of ET-STATCOM; k_s is defined as an inertia scaling factor.

Substituting (7) into (4), we have

$$k_s T_{sc} \frac{d\omega_i}{\omega_{g0} dt} = -\frac{p_s}{P_0}. \quad (8)$$

Comparing (5) with (8), it can be seen that the MC-based energy synchronization link shows similar active power-frequency characteristics with the SG. In (8), the set of $k_s T_{sc}$ is equivalent to the inertia constant T_{sg} of SG. Obviously, the parameter k_s can be used to flexibly adjust the virtual inertia magnitude of the SCs to provide an inertia response capability beyond that of the SG.

According to (6), the small-signal model of the active power p_s can be expressed as

$$\Delta p_s = \frac{3U_{gm0}U_{m0}}{2X_g} \Delta \delta = \frac{3U_{gm0}U_{m0}}{2X_g} \frac{1}{s} (\Delta \omega_i - \Delta \omega_g) \quad (9)$$

where U_{gm0} and U_{m0} denote the steady-state values of U_{gm} and U_m , respectively.

As depicted in Fig. 7, assuming a sudden load increase, the grid frequency ω_g will drop due to the power deficit, causing the power angle δ to widen. As per (9), the ET-STATCOM increases its active power output, achieving an initiative inertia response. Consequently, based on (4), the enhanced active power prompts the SCs to discharge, reducing their stored energy w_{ts} . By properly setting the coefficient k_s in accordance with grid frequency stability constraints (as analyzed in Section IV-B), the SC energy charging/discharging limits can be effectively regulated. Moreover, (7) shows that the decrease in SC energy lowers the ET-STATCOM's output frequency until it resynchronizes with the grid frequency ω_g .

B. APDF-Based Damping Enhancement Technique

When employing the SC energy dynamics directly for synchronization, the system encounters an undamped oscillation issue. However, as previously analyzed, none of the existing MC damping enhancement techniques are suitable for the ET-STATCOM.

To solve the above-mentioned problem, an active power output-based damping term can be added to the synchronization

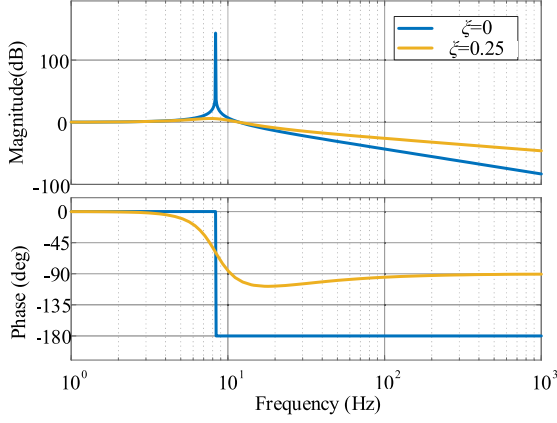


Fig. 8. Bode diagram of $G_{\omega}(s)$ before and after adding damping term.

control link. This is done by amending (7) as follows:

$$\frac{w_{ts} - W_{ts0}}{2W_{ts0}} - \frac{D}{P_0 T_{sc}} p_s = k_s \frac{\omega_i - \omega_{g0}}{\omega_{g0}} \quad (10)$$

where D is defined as a damping coefficient.

Substituting (10) into (4), we can obtain

$$k_s T_{sc} \frac{d\omega_i}{\omega_{g0} dt} = -\frac{p_s}{P_0} - \frac{D dp_s}{P_0 dt}. \quad (11)$$

Linearizing (11), yields

$$\Delta\omega_i = -\frac{(1 + Ds)\omega_{g0}}{k_s T_{sc} P_0 s} \Delta p_s. \quad (12)$$

Then, the closed-loop transfer function between $\Delta\omega_i$ and $\Delta\omega_g$ can be expressed as

$$G_{\omega}(s) = \frac{\Delta\omega_i}{\Delta\omega_g} = \frac{DKs + K}{s^2 + DKs + K} = \frac{2\xi\omega_n s + \omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad (13)$$

where

$$\begin{cases} K = \frac{3U_{gm0}U_{m0}\omega_{g0}}{2X_g k_s T_{sc} P_0} \\ \omega_n = \sqrt{K} \\ \xi = \frac{D\sqrt{K}}{2} \end{cases}. \quad (14)$$

Equation (14) defines ξ as the damping ratio and ω_n as the natural oscillation frequency. Fig. 8 presents the Bode diagram of $G_{\omega}(s)$, comparing systems with and without the proposed damping enhancement strategy. The case where $\xi = 0$ represents the closed-loop transfer function from (7). The results demonstrate that the damping compensation strategy effectively eliminates the resonance peak in synchronization control, thereby enhancing both synchronization stability and dynamic performance. More importantly, this damping enhancement method maintains consistent performance across varying active power levels and operates independently of PLL, offering superior overall performance.

According to (13), the closed-loop error transfer function $E_{\omega}(s)$ can be further deduced as

$$E_{\omega}(s) = 1 - G_{\omega}(s) = \frac{s^2}{s^2 + DKs + K}. \quad (15)$$

Assuming the grid frequency ω_g experiences a step drop, we can express the frequency deviation as $\Delta\omega_g(t) = -\varepsilon(t)$. In the Laplace domain, this becomes $\Delta\omega_g(s) = -1/s$. Applying the Final Value Theorem of Laplace transform yields

$$E_{\omega}(\infty) = \lim_{s \rightarrow 0} s E_{\omega}(s) \frac{1}{s} = 0. \quad (16)$$

Equation (16) indicates that the output frequency of the ET-STATCOM can track the grid frequency without static error. Hence, the proposed damping enhancement scheme does not alter the self-synchronization characteristics of the energy synchronization link.

Moreover, the transfer function of the active power output with respect to the grid frequency is

$$\begin{aligned} \frac{\Delta p_s(s)}{\Delta\omega_g(s)} = & -\frac{DKk_s T_{sc} P_0 s^2 + Kk_s T_{sc} P_0 s}{D\omega_{g0} s^3 + \omega_{g0}(1 + D^2K)s^2 + 2DK\omega_{g0}s + K\omega_{g0}}. \end{aligned} \quad (17)$$

Supposing that the grid frequency ω_g is step-dropped, the steady-state value of output active, i.e., $\Delta p_s(\infty)$, is derived as

$$\Delta p_s(\infty) = \lim_{s \rightarrow 0} s \Delta p_s(s) \frac{-1}{s} = 0. \quad (18)$$

Equation (18) shows that the system automatically terminates active power output upon completing the required inertial response, preventing SCs from engaging in primary frequency regulation. This design better aligns with the energy characteristics of SCs.

Next, the transfer function between Δw_{ts} and $\Delta\omega_g$ can be expressed as

$$\frac{\Delta w_{ts}(s)}{\Delta\omega_g(s)} = \frac{\Delta w_{ts}(s)}{\Delta\omega_i(s)} \frac{\Delta\omega_i(s)}{\Delta\omega_g(s)} = \frac{2k_s W_{ts0}}{(1 + Ds)\omega_{g0}} \frac{DKs + K}{s^2 + DKs + K}. \quad (19)$$

Supposing that the grid frequency ω_g is step-dropped, the variation of the SC energy can be described in Laplace form as

$$\Delta w_{ts} = \frac{2k_s W_{ts0}}{(1 + Ds)\omega_{g0}} \frac{DKs + K}{s^2 + DKs + K} \frac{1}{s}. \quad (20)$$

According to the Final-value theorem of the Laplace transform, we have

$$\Delta w_{ts}(\infty) = \lim_{s \rightarrow 0} s \Delta w_{ts}(s) \frac{1}{s} = \frac{2k_s W_{ts0}}{\omega_{g0}}. \quad (21)$$

Equation (21) shows that the SC energy changes by $2k_s\%$ for a 1% change in the grid frequency. In other words, the SC energy will follow the movement of the grid frequency. When the grid frequency is forced to be set within a specific range, the variable k_s can be adjusted to ensure that the SC energy is within its reasonable range. Therefore, no additional control links are required to monitor the SC energy anymore, resulting in reduced control complexity. Besides, (21) demonstrates that the SC energy variation depends solely on k_s and remains independent of D . In other words, the proposed damping enhancement scheme does not alter the SC energy distribution characteristics of the energy synchronization link.

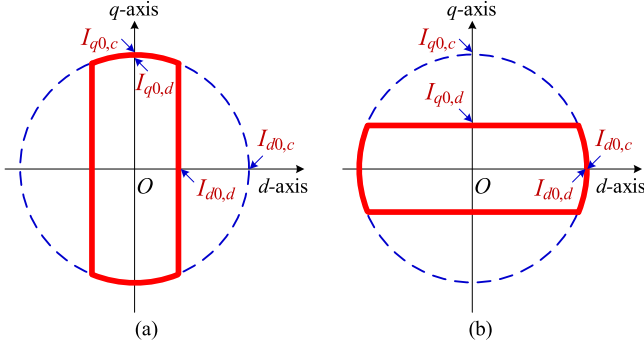


Fig. 9. Schematic illustration of current limiting requirements for different design conditions. (a) $P_0 < Q_0$ condition. (b) $P_0 > Q_0$ condition.

To sum up, the proposed damping enhancement scheme is independent of the active power level of the equipment and does not involve PLL. Moreover, it does not alter the self-synchronization characteristics, inertial response characteristics, and SC energy distribution characteristics of the energy synchronization link. This technology enables ET-STATCOM to integrate PLL-less self-synchronization, power oscillation suppression, spontaneous inertial response, and autonomous management of SC energy.

C. MCLC Technique

It is necessary to design a current limiting controller for converters and SCs to avoid over-current. In fact, a widely used circular current limiter (CCL) has been proposed in [34] to limit the current and maintain the GFMC characteristics during grid faults. However, it should be noted that using this technique in the ET-STATCOM may cause SC overcurrent during transients when the active power capacity of the equipment is much less than its reactive power capacity.

The generalized rated values of the three-phase grid currents in the d - q frame, i.e., are approximately as

$$I_{d0} = \frac{P_0}{1.5U_{m0}}, \quad I_{q0} = \frac{Q_0}{1.5U_{m0}}. \quad (22)$$

It is noteworthy that P_0 and Q_0 have been determined during the design stage of the equipment and are independent of its operational state. As analyzed previously, P_0 and Q_0 are not necessarily equal for the ET-STATCOM due to the decoupling characteristics of active and reactive power flows. Hence, we can define $I_{d0,c}$ and $I_{q0,c}$ as the rated values of the three-phase grid currents in the d - q frame when $P_0 = Q_0$, and $I_{d0,d}$ and $I_{q0,d}$ as the rated values of the three-phase grid currents in the d - q frame when $P_0 \neq Q_0$. Obviously, for the equipment already in operation or to be commissioned, only one set of parameters ($I_{d0,c}$, $I_{q0,c}$ or $I_{d0,d}$, $I_{q0,d}$) can be selected.

According to the above definitions, when $P_0 = Q_0$, one has

$$I_{lim} \approx I_{d0,c} = I_{q0,c} = \frac{P_0}{1.5U_{m0}} = \frac{Q_0}{1.5U_{m0}} \quad (23)$$

where I_{lim} denotes the current limit value of the equipment.

In this case, the equipment's safe area is a circle, as shown by the area enclosed by the blue dotted line in Fig. 9. The circle's

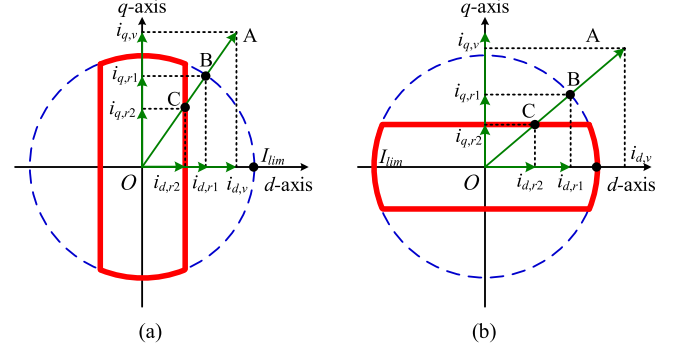


Fig. 10. Schematic illustration of the proposed MCLC for different design conditions. (a) $P_0 < Q_0$ condition. (b) $P_0 > Q_0$ condition.

radius is I_{lim} . Obviously, the CCL can limit the operating area of the equipment to the circle.

When $P_0 \neq Q_0$, two cases need to be discussed, namely

- 1) $P_0 < Q_0$, then $I_{lim} \approx I_{q0,d} > I_{d0,d}$ can be obtained.
- 2) $P_0 > Q_0$, then $I_{lim} \approx I_{d0,d} > I_{q0,d}$ can be obtained.

In such cases, the safe operating area of the equipment is reduced to the region enclosed by the red solid line, as shown in Fig. 9. Clearly, this red-bounded area represents a subspace of the original circular region, demonstrating that conventional CCL may not adequately protect the SCs.

To avoid this drawback, this article proposes an MCLC technique. It reuses the classical CCL in the first stage, i.e.,

$$i_{dq,r1} = \begin{cases} i_{dq,v} & \|i_{dq,v}\| \leq I_{lim} \\ \frac{I_{lim}}{\|i_{dq,v}\|} i_{dq,v} & \|i_{dq,v}\| > I_{lim} \end{cases} \quad (24)$$

where $i_{dq,r1}$ denote the outputs of CCL; $i_{dq,v}$ are the outputs of virtual admittance control (VAC) described below.

With the help of the first stage, the ET-STATCOM's operating point will move from point A to point B, as shown in Fig. 10. Then, to avoid an over-current of SCs, the operating point must also be moved from B to C, which can be achieved by feeding $i_{dq,r1}$ into the second current limiting link. This is achieved by

$$i_{dq,r2} = \begin{cases} i_{dq,r1} & \|i_{dq,r1}\| \leq \|i_{d0,d}\| \\ i_{dq,r1} \frac{i_{d0,d}}{i_{d,r1}} & \|i_{dq,r1}\| > \|i_{d0,d}\| \end{cases} \quad \text{for } P_0 \leq Q_0$$

$$i_{dq,r2} = \begin{cases} i_{dq,r1} & \|i_{dq,r1}\| \leq \|i_{q0,d}\| \\ i_{dq,r1} \frac{i_{q0,d}}{i_{d,r1}} & \|i_{dq,r1}\| > \|i_{q0,d}\| \end{cases} \quad \text{for } Q_0 \leq P_0. \quad (25)$$

Obviously, the proposed MCLC technique can maximize the reactive power capability of the ET-STATCOM while ensuring the safety of its SCs. In addition, when $P_0 = Q_0$, i.e., $I_{d0,d} = I_{q0,d}$, $i_{dq,r2} = i_{dq,r1}$ can be obtained, meaning that the technique is adaptively compatible with both $P_0 = Q_0$ and $P_0 \neq Q_0$ conditions.

D. Overall Architecture of the Improved Grid Forming Control Technique

The overall control structure of the ET-STATCOM with the proposed improved GFMC is presented in Fig. 11. The output

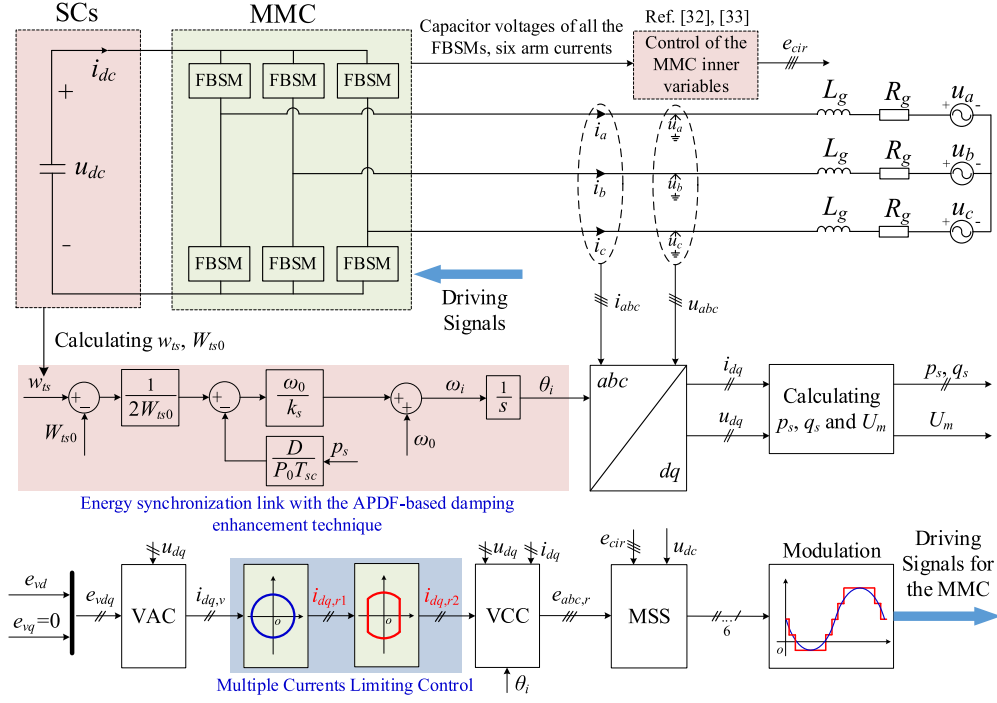


Fig. 11. Overall architecture of the proposed improved GFMC for the ET-STATCOM.

of the synchronous link is the angle θ_i , which is used to perform coordinate transformation. The VAC is used to enhance grid impedance characteristics [35]. The basic control scheme of the VAC is

$$\begin{cases} i_{d,v} = \frac{e_{vd} - u_d + \omega_i L_v i_{q,c}}{sL_v + R_v} \\ i_{q,v} = \frac{e_{vq} - u_q - \omega_i L_v i_{d,c}}{sL_v + R_v} \end{cases} \quad (26)$$

where R_v is the virtual resistance, and L_v is the virtual inductance; e_{vdq} , u_{dq} , and $i_{dq,v}$ are the virtual voltage, PCC voltage, and current command in the d - q frame, respectively.

Next, the outputs of the VAC are fed into the MCLC, which generates the ultimate current reference in the d - q frame. Then, the conventional vector current control (VCC) is adopted to create the ac inner voltage. Finally, the driving signals of the 6N power transistors of the MMC are generated by modulation signal synthesis and PWM modulation. The design of the VAC and VCC is the same as that in some existing GFMCs [36], [37].

IV. SMALL SIGNAL MODELING AND CONTROL DESIGN

A. Small Signal Modeling

The state space model of the ET-STATCOM with the proposed improved GFMC is built to investigate the small-signal stability of the system. It is worth noting that only the control links shown in Fig. 11 are involved. Linearizing the control links in Fig. 11 at the steady state operating point, the small-signal model of the system in the d - q frame can be established. Due to the article's page limit, the exact derivation process will not be presented. The resulting small signal model is given by

$$d\Delta x/dt = A\Delta x \quad (27)$$

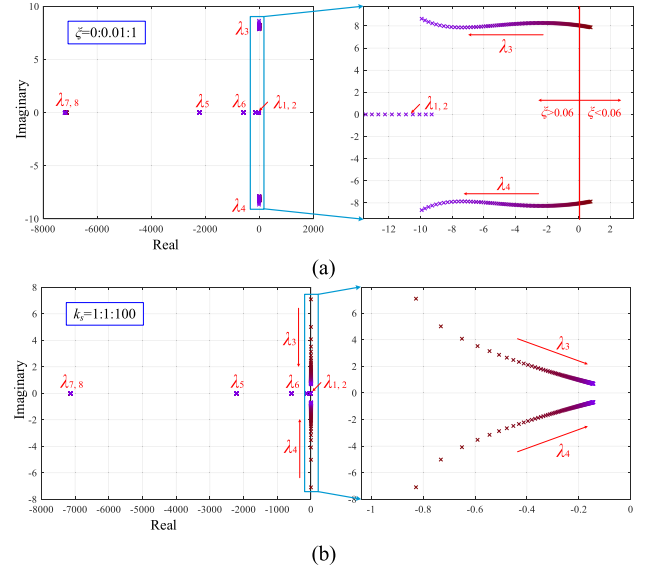


Fig. 12. Eigenvalue distributions of the ET-STATCOM when using the proposed method with the variations of ξ and k_s . (a) ξ varies. (b) k_s varies.

where $\Delta x = [\Delta w_{ts}, \Delta p_s, \Delta \omega_i, \Delta \delta, \Delta i_{d,v}, \Delta i_{q,v}, \Delta i_d, \Delta i_q, \Delta x_d, \Delta x_q]$.

The variables Δx_d and Δx_q in (27) represent the outputs of the integrators of the VCC. The expression for Matrix A is shown in Appendix A.

Fig. 12 presents the eigenvalue variations with ξ and k_s using the parameters in Appendix B, with arrows showing their trajectory upon parameter increase.

Fig. 12(a) gives the results when ξ varies. The real parts of the obtained eigenvalues are negative when $\xi > 0.06$, proving that the

control system is stable. However, when $\xi < 0.06$, the eigenvalues λ_1 and λ_2 move to the right half plane, implying that the system is destabilized. Hence, $\xi > 0.06$ must be satisfied to maintain the system's stability.

Fig. 12(b) presents the results for varying k_s . The obtained eigenvalues consistently exhibit negative real parts across all k_s values, demonstrating that the system stability remains unaffected by changes in k_s .

B. Design Consideration for K_s and D

Since k_s have no impact on the system's stability, its constraints should be considered regarding inertia response capability and the variation range of the SC energy.

For one thing, as seen from (8) and (11), k_s must be greater than 1 not to lose the inertia response capability of the VSM. For another, considering the limited variation range of SC energy, k_s must also have an upper limit.

The expression for the release degree of SC energy can be written as

$$w_{re} = \frac{\frac{1}{2}C_{se}U_{dc0}^2 - \frac{1}{2}C_{se}(\lambda U_{dc0})^2}{\frac{1}{2}C_{se}U_{dc0}^2} = 1 - \lambda^2 \quad (28)$$

where w_{re} denotes the energy magnitude released by SCs; λ is defined as the ratio between the minimum voltage and the rated voltage of SCs, and it satisfies $\lambda \in [0, 1]$.

As shown in (28), when $\lambda = 1/2$, $w_{re} = 0.75$ can be obtained, meaning that 75% of the SC energy has been released when its voltage is discharged from its rated value to one-half of its rated value. Although the theoretical minimum voltage of SCs is zero, a margin has to be retained to avoid extreme operating conditions of the power grid [38]. Consequently, it makes more sense to set λ to 1/2.

Without loss of generality, we define ω_m ($\omega_m > 0$) as the maximum deviation allowed for the grid frequency, then k_s can be determined as

$$k_s = \frac{1}{2} \left| \frac{w_{ts,r} - W_{ts0}}{W_{ts0}} \right|_{\max} \left| \frac{\omega_0}{\omega_g - \omega_0} \right|_{\max} = 0.375 \frac{\omega_{g0}}{\omega_m}. \quad (29)$$

In mainstream international grid frequency standards, the maximum allowable frequency deviation is typically mandated at ± 0.5 Hz [39], [40]. Consequently, the upper limit for the parameter k_s should not exceed 37.5.

In summary, the range of values of k_s is

$$1 \leq k_s \leq 37.5. \quad (30)$$

Hence, to maximize the inertial response capability of the ET-STATCOM, $k_s = 37.5$ should be taken.

Substituting $k_s = 37.5$ into (7) yields

$$w_{ts} = \left(0.75 \frac{\omega_i - \omega_{g0}}{\omega_m} + 1 \right) W_{ts0}. \quad (31)$$

We can draw the following conclusions based on (31).

- 1) When $\omega_g = \omega_{g0} - \omega_m$, $w_{ts} = 0.25W_{ts0}$ can be got, meaning that the SC voltage drops from the rated value U_{dc0} to $0.5U_{dc0}$.

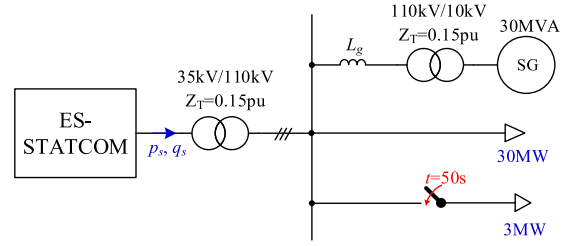


Fig. 13. System structure diagram of the inertia response verification.

- 2) When $\omega_g = \omega_{g0} + \omega_m$, $w_{ts} = 1.75W_{ts0}$ can be got, meaning that the SC voltage rises from the rated value U_{dc0} to $1.32U_{dc0}$.

As a result, when the grid frequency is distributed in the range of $[\omega_{g0} - \omega_m, \omega_{g0} + \omega_m]$, the SC voltage will follow this distribution in the range of $[0.5U_{dc0}, 1.32U_{dc0}]$, which realizes the autonomous control of the SC voltage and energy.

The value of variable D is mainly constrained by its impact on system stability. As indicated in (14), a smaller grid SCR requires a larger minimum value of D . Therefore, to ensure system stability across a wide range of SCRs, the selection of D must satisfy this condition

$$D = \frac{2\xi}{\sqrt{K}} \geq 2\xi_m \sqrt{k_s T_{sc}} \quad (32)$$

where ξ_m denotes the minimal damping ratio for system stabilization. In this article, $\xi_m = 0.06$.

V. SIMULATION ANALYSIS

A 35kV/30MVar/3MW/4s ET-STATCOM system is modeled with the professional tool PSCAD/EMTDC to verify the proposed control strategy. The system simulation parameters are gathered in Appendix B.

A. Simulation Results of Inertia Response Capability

First, the effectiveness and advancement of the proposed technique are illustrated by the comparative simulations of different synchronization and damping enhancement control strategies. The following simulation cases are set up.

Case I: Using the VSM.

Case II: Using the cascade-type modified VSM, as shown in Fig. 2(b).

Case III: Using the MC with the GSC voltage output modification-based damping enhancement technique, as shown in Fig. 4(b).

Case IV: Using the energy synchronization link with the APDF-based damping enhancement technique, as shown in Fig. 11.

A sudden increase in load is simulated as an example. The structure diagram of the simulation system is shown in Fig. 13. The simulation procedure is set up as follows: a 30-MW load is supplied by the SG before $t = 50$ s while the ET-STATCOM is connected to the grid without power output. When $t = 50$ s, an additional 3 MW load is put in. The control group is the case where the ET-STATCOM is not placed.

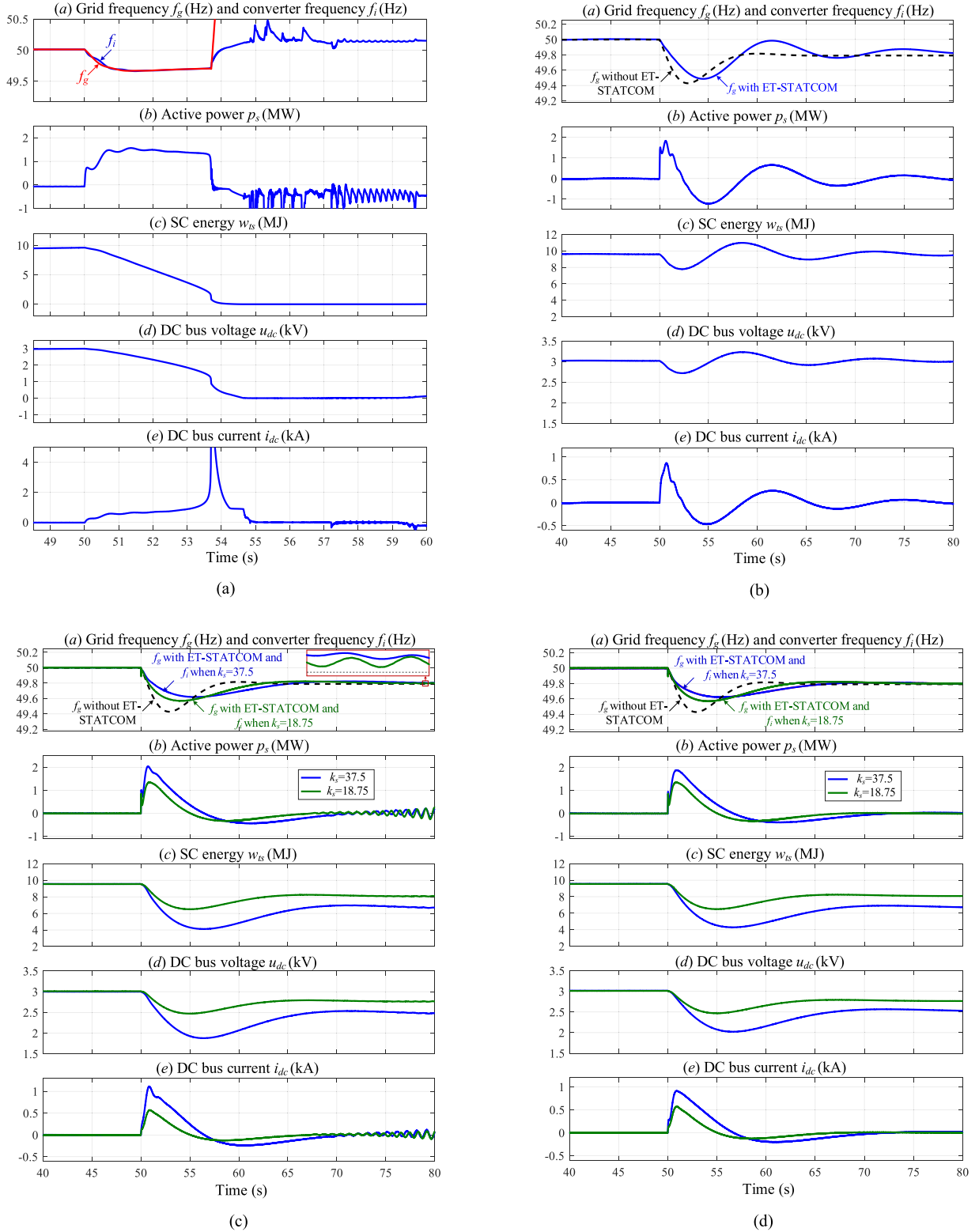


Fig. 14. Simulation results for a sudden load increase with different control techniques. (a) Case I. (b) Case II. (c) Case III. (d) Case IV.

Fig. 14(a) shows the simulation results of Case I, where the virtual inertia time constant $J = 4$ s and virtual damping coefficient $D = 20$ are set [19], [35]. As indicated, the system can operate stably up to 53.7 s. However, it starts to diverge after $t = 53.7$ s when the SC voltage and energy rush to zero due to the absence of SC voltage in the synchronous control loop. It

is worth noting that setting the damping coefficient D to zero directly after the SC voltage reaches zero will not work either since the system will oscillate without damping.

Fig. 14(b) presents the simulation results of Case II, where $J = 4$ s and $D = 20$ are set, and the proportional and integral coefficients are set to be 0.5 and 2, respectively. These control

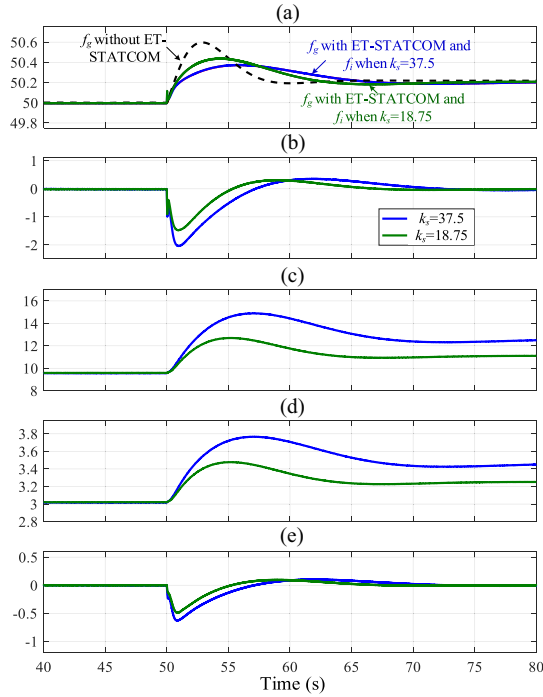


Fig. 15. Simulation results for a sudden load decrease condition with the proposed technique. (a) Grid frequency f_g (Hz) and converter frequency f_i (Hz). (b) Active power p_s (MW). (c) SC energy w_{ts} (MJ). (d) DC bus voltage u_{dc} (kV). (e) DC bus current i_{dc} (kA).

parameters were derived through theoretical analysis based on [41] combined with extensive trial-and-error tuning to maintain the SC energy within allowable limits. The results indicate that the ET-STATCOM can deliver active power to provide inertia response during frequency drops. However, in the new steady state, the SC energy returns to its rated value despite the sustained grid frequency deviation, indicating zero net energy contribution from the SCs. Consequently, this method exhibits inferior frequency support capability compared to Fig. 14(d). While increasing D enhances the inertial response, it also increases the risk of SC energy depletion. Furthermore, transient oscillations are observed due to cross-coupling between the SC energy and active power control loops.

Fig. 14(c) shows the simulation results of Case III. When the grid frequency drops, the ET-STATCOM can output some active power to reduce the frequency deviation. At the end of the inertial response, the ET-STATCOM stops outputting active power. However, in the new steady state, the active power and frequency of the ET-STATCOM begin to oscillate at low frequencies. This is because its damping enhancement technique depends on the active power level and provides zero damping at a low active power level.

Fig. 14(d) shows the simulation results of Case IV. Obviously, the system can be stable all the time. When $k_s = 37.5$ and $k_s = 18.75$, the ET-STATCOM emits a maximum of 1.92 MW and 1.39 MW of active power, which helps to raise the minimum value of SG frequency from 49.43 to 49.63 Hz and 49.58 Hz, respectively. Also, the rates of change of frequency (RoCoF) are both improved. As shown in the first subplot of Fig. 14(d), the converter frequency f_i can closely follow the SG frequency both for $k_s = 37.5$ or $k_s = 18.75$.

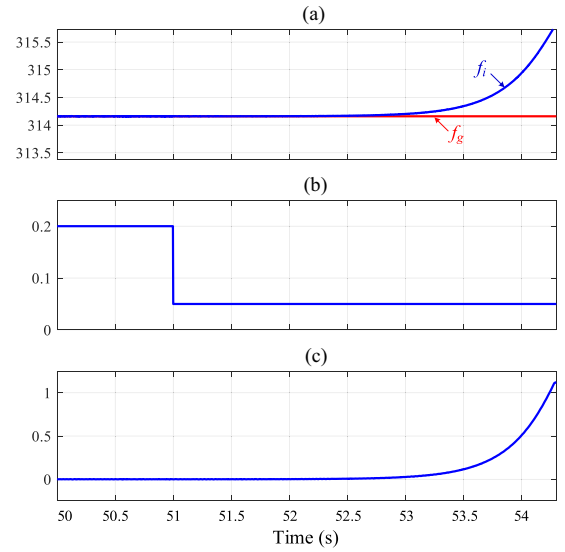


Fig. 16. Simulation results of system destabilization due to the decrease in the damping ratio of the proposed synchronization link. (a) Grid frequency f_g (Hz) and converter frequency f_i (Hz). (b) Damping ratio ξ . (c) DC bus current i_{dc} (kA).

In the case of $k_s = 37.5$, when the SG frequency reaches 49.63 Hz and 49.80 Hz, the SC energy arrives at 4.31 MJ (Theoretical value is 4.25 MJ) and 6.75 MJ (Theoretical value is 6.69 MJ), respectively. In the case of $k_s = 18.75$, when the SG frequency reaches 49.57 Hz and 49.80 Hz, the SC energy arrives at 6.59 MJ (Theoretical value is 6.51 MJ) and 8.09 MJ (Theoretical value is 8.11 MJ), respectively. The above-mentioned results are consistent with the theoretical calculations (31). These results signify that the initiative inertia response and the autonomous control of SC energy (voltage) are creatively integrated with the proposed synchronization link. Besides, a larger k_s can enhance the inertial inertia response capability of the ET-STATCOM without compromising system stability.

Fig. 15 depicts the simulation results using the proposed technique (The method given in Case IV) with a sudden load reduction condition. The simulation system's structure diagram is similar to Fig. 13. The simulation step is set up as follows: a 30 MW load is supplied by the SG before $t = 50$ s, and then a 3-MW load is cut out when $t = 50$ s.

As can be seen, the ET-STATCOM assimilates a maximum of 2 MW and 1.43 MW of active power when $k_s = 37.5$ and $k_s = 18.75$, respectively, which contributes to the improvement of the frequency stability, both in terms of RoCoF and maximum frequency values. In the case of $k_s = 37.5$, when the SG frequency reaches 50.37 Hz and 50.21 Hz, the SC energy arrives at 14.73 MJ (Theoretical value is 14.82 MJ) and 12.43 MJ (Theoretical value is 12.55 MJ), respectively. In the case of $k_s = 18.75$, when the SG frequency reaches 50.44 Hz and 50.21 Hz, the SC energy arrives at 12.65 MJ (Theoretical value is 12.70 MJ) and 11.10 MJ (Theoretical value is 11.05 MJ), respectively. Obviously, the results indicate that the stability of SC energy and voltage is maintained while empowering the ET-STATCOM initiative inertia response.

In addition, Fig. 16 gives the simulation results when the damping ratio ξ is reduced from 0.2 to 0.05. It can be seen that

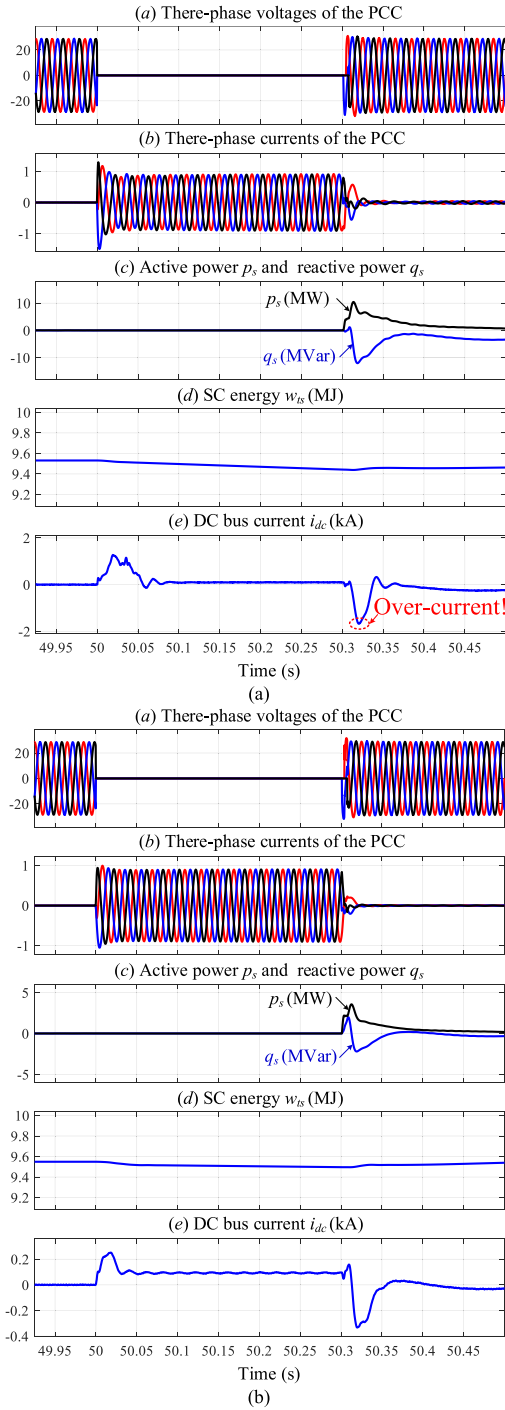


Fig. 17. Simulation results during zero-voltage fault with the two different current limiting techniques. (a) CCL. (b) MCLC.

the system is gradually destabilized after the damping ratio ξ is reduced from 0.2 to 0.05, which indicates the effectiveness of the damping enhancement technique proposed in this article. The simulation results in Fig. 16 are consistent with the conclusion of the stability analysis in Fig. 12(a).

B. Simulation Results of Voltage Support Capability

The voltage support capability simulation steps are set as follows: a 30-MW load is supplied by the SG before $t = 30$ s. When $t = 30$ s, a three-phase zero-voltage fault with a duration of

0.3 s occurs in the PCC. Fig. 17(a) and (b) shows the simulation results with the proposed energy synchronization control as well as the conventional CCL and the proposed MCLC technique, respectively.

As depicted in Fig. 17(a), the SCs reached a current of 1.55 kA during grid restoration, exceeding its tolerance. Fortunately, this phenomenon is avoided when using the proposed MCLC technique, as shown in Fig. 17(b). This is because the proposed strategy considers the difference between active and reactive power in advance and enforces a separate active/reactive current limiting.

Furthermore, the transient behaviors of the proposed control strategies under grid faults should also be explored. Transient stability is highly dependent on factors such as fault location, fault severity, and fault duration (FD). To thoroughly evaluate the transient stability performance of the proposed control strategies, a large number of time-domain simulations are performed considering different fault locations, grounding resistances or grid voltage drops, and FDs, as illustrated in Fig. 18. These figures respectively represent three-phase low-voltage faults occurring at: (1) the PCC, (2) the midpoint of the grid impedance, and (3) the power grid.

At each fault location, the FD is systematically varied across 0.2, 0.5, 1, 3, 5, 7, and 9 s to assess the system's transient stability. The figures present key dynamic responses, including

- 1) the power angle δ (phase difference between the PCC voltage and grid voltage),
- 2) the SC energy, and
- 3) the fault current magnitude of the ET-STATCOM generated during the fault period.

Lines of the same color indicate different variables for the same operating condition.

Fig. 18 illustrates that the ET-STATCOM generates varying short-circuit currents during three distinct grid fault conditions. This current injection contributes to power losses in both the semiconductor switches and internal lines, thereby depleting the stored energy in the SC. Simultaneously, the interaction between the ET-STATCOM, the faulted line, and the power grid induces active and reactive power redistribution governed by power angle and voltage amplitude characteristics. This power exchange may either dissipate energy from or replenish energy to the SC. As a consequence of these dynamic interactions, the output frequency of the ET-STATCOM diverges from the grid frequency, inspiring a drawn-out power angle.

As the fault persists, the power angle progressively increases. Once it surpasses the unstable equilibrium point, system instability occurs, as depicted in Fig. 18(a). Here, a three-phase-to-ground fault at the PCC triggers maximum reactive current injection from the ET-STATCOM, causing the highest internal losses and the fastest SC energy depletion. Hence, the power angle expands most rapidly. Fortunately, even under PCC zero-voltage faults, the system destabilizes only after a prolonged duration of 9 s—far exceeding the typical critical clearing time of approximately 200 ms [42], [43]. Besides, the ET-STATCOM remained stable over an extended period when the faults occurred at the midpoint of the grid impedance and at the power grid. These results indicate that the proposed control strategies exhibit robust transient stability margins.

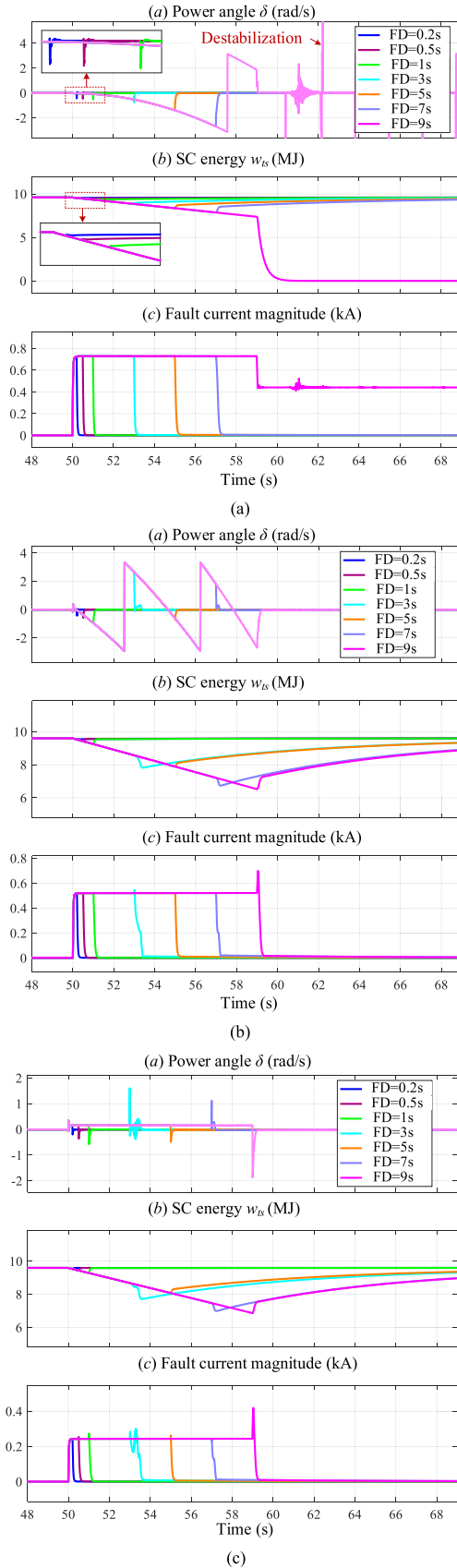


Fig. 18. Simulation results of three-phase low-voltage faults at different locations. (a) PCC. (b) Midpoint of the grid impedance. (c) Power grid.

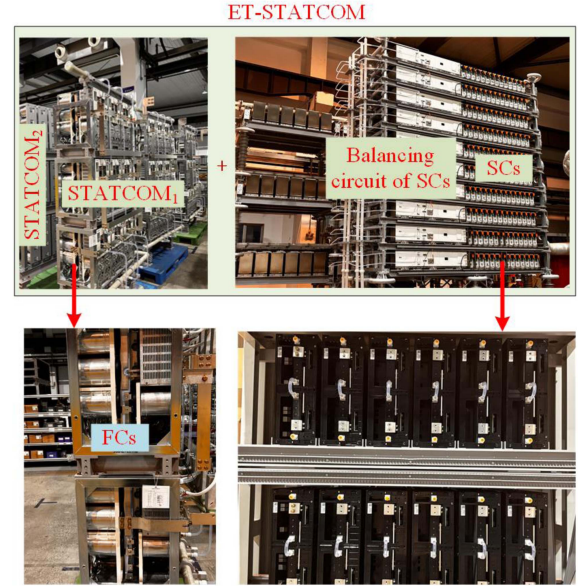


Fig. 19. Internal arrangement of a practical 35kV/30MVar/3MW/4s ET-STATCOM prototype.

VI. EXPERIMENTAL VERIFICATION

Due to the potential of the ET-STATCOM to contribute to grid frequency and voltage stabilization, a practical 35kV/30MVar/3MW/4s ET-STATCOM prototype has been manufactured. The internal arrangement of the equipment is depicted in Fig. 19.

A. Experimental Results of Inertia Response Capability

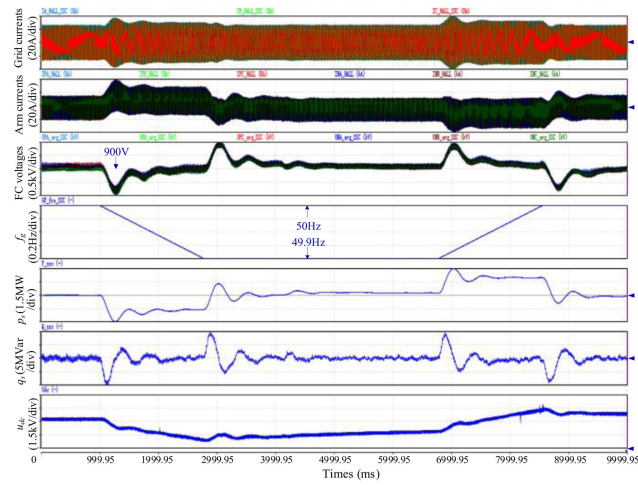
A 5-MW-rated grid simulator was employed to test the inertial response capability. The test results are presented in Fig. 20(a) and (b), demonstrating the frequency response under downward and upward grid frequency variations, respectively.

The seven subgraphs from top to bottom are as follows:

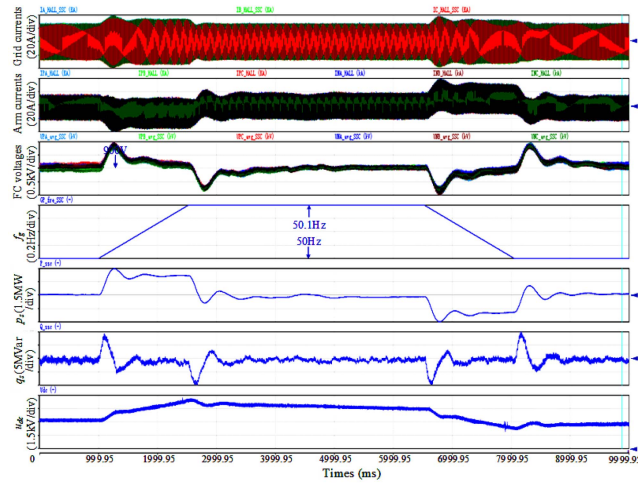
- 1) Three-phase currents at the PCC (20 A per grid).
- 2) Six-arm currents of the MMC (20 A per grid).
- 3) Capacitor voltages of the first FBSM in each arm of the MMC.
- 4) Frequency of the ac power grid (0.2 Hz per grid).
- 5) Active power at the PCC (1.5 MW per grid).
- 6) Reactive power at the PCC (5 MVar per grid).
- 7) DC bus voltage of the MMC (1.5 kV per grid).

The right triangle of each subgraph indicates its zero position. The right triangle in each subplot marks the zero-reference position.

As illustrated in Fig. 20, the ET-STATCOM demonstrates the capability to inject or absorb a controlled amount of active power within the safe operating limits of the SC voltage and current during grid frequency deviations. In addition, when using the proposed control technique, there are no over-voltage or over-current issues, and the inertial response performance is satisfactory.



(a)



(b)

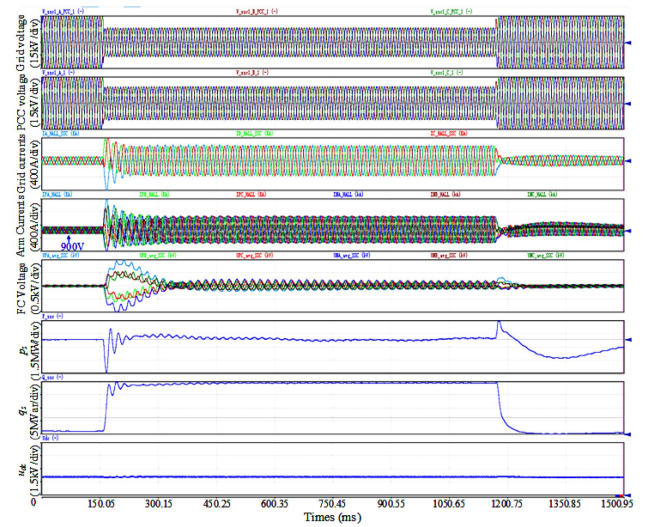
Fig. 20. Experimental waveforms when the grid frequency varies. (a) Grid frequency varies downward. (b) Grid frequency varies upward.

B. Experimental Results of Voltage-Supporting Capability

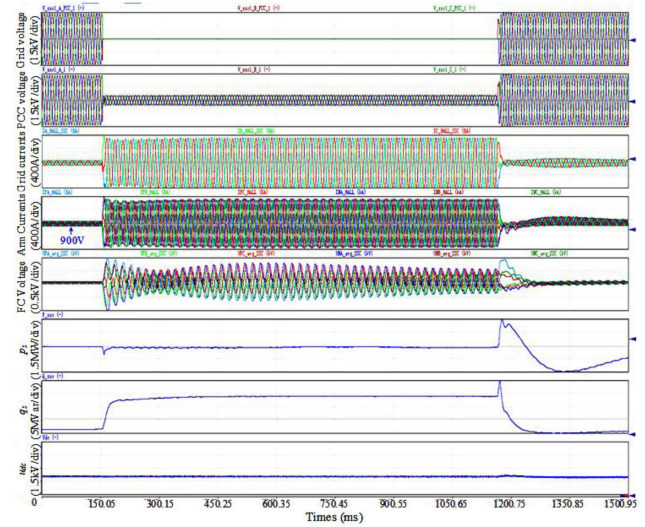
Experimental results for the verification of voltage-supporting capability are shown in Fig. 21. The eight subgraphs from top to bottom are as follows:

- 1) Three-phase grid voltages (15 kV per grid).
- 2) Three-phase voltages at the PCC (15 kV per grid).
- 3) Three-phase currents at the PCC (20 A per grid).
- 4) Six-arm currents of the MMC (20 A per grid).
- 5) Capacitor voltages of the first FBSM in each arm of the MMC (0.5 kV per grid).
- 6) Active power at the PCC (1.5 MW per grid).
- 7) Reactive power at the PCC (5 MVar per grid).
- 8) DC bus voltage of the MMC (1.5 kV per grid).

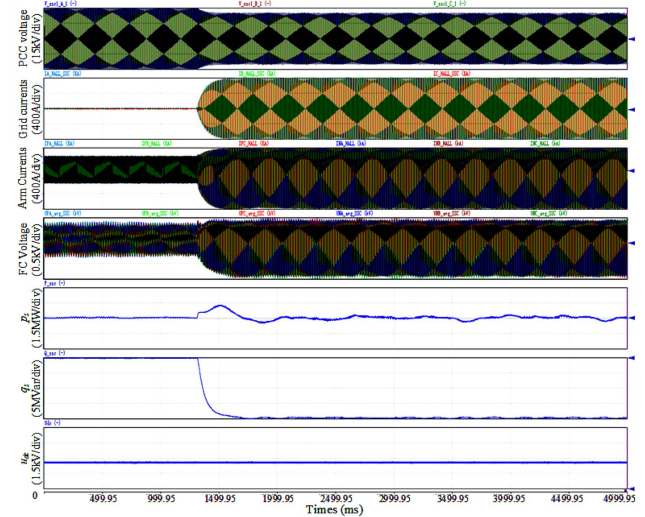
1) *Grid Voltage Drops to 0.5 pu*: Fig. 21(a) shows the experimental waveforms when the grid voltages drop to 0.5 pu. It can be seen that the reactive power increases rapidly to boost the PCC voltage to 0.6 pu during the low-voltage fault. The SCs and



(a)



(b)



(c)

Fig. 21. Experimental results of reactive power response capability. (a) 0.5 pu-low voltage fault. (b) Zero-voltage fault. (c) Reactive power command varies.

$$A = \begin{bmatrix} 0 & -\frac{2W_{ts0}}{T_{sc}P_0} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3U_{gm0}U_{m0}}{2X_g} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\omega_{g0}}{k_s T_{sc} P_0} & -\frac{D\omega_{g0}}{k_s T_{sc} P_0} \frac{3U_{gm0}U_{m0}}{2X_g} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \omega_{g0} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{R_v}{L_v} & -\omega_i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \omega_i & -\frac{R_v}{L_v} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{k_p}{L_{ae}} & 0 & -\frac{k_p+R_{ae}}{L_{ae}} & 0 & \frac{k_i}{L_{ae}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{k_p}{L_{ae}} & 0 & -\frac{k_p+R_{ae}}{L_{ae}} & 0 & \frac{k_i}{L_{ae}} \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 \end{bmatrix}.$$

FCs do not suffer from over-current and over-voltage problems during transients.

2) *Grid Voltage Drops to 0*: As shown in Fig. 21(b), after the occurrence of zero voltage fault, the reactive current rises rapidly, and the PCC voltage amplitude is elevated to 0.15 pu, which demonstrates the effectiveness of the proposed method in terms of low voltage support capability. Similarly, SCs and FCs are free from over-current and over-voltage problems during transients.

3) *The Reactive Power Setting Value Varies*: The reactive power setting value $Q_{s,r}$ steps from zero to -20 Mvar. Fig. 21(c) shows the experimental waveforms in this case. When the controller receives a reactive power command, the ET-STATCOM outputs a reactive power magnitude that matches the command. Zero active power fluctuations characterize the process, implying that SCs and FCs are safe enough.

VII. CONCLUSION

The ET-STATCOM has been deemed as one of the key stabilizers for the modern power grid. However, its proper GFMC remains undeveloped, mainly because:

- 1) the classical VSM does not possess SC voltage management capability,
- 2) the modified FC-STATCOM-oriented VSMs lack inertial response capability,
- 3) the damping enhancement method in the MC's dc voltage control loop fails for the ET-STATCOM, and
- 4) the existing current limiting strategies for the GFMC can only limit the total current amplitude without the limiting capability of active and reactive currents.

To avoid these drawbacks, an improved GFMC for the ET-STATCOM is proposed in this article, including the MC-based energy synchronization link, the APDF-based damping enhancement scheme, and the MCLC technique. The proposed damping enhancement scheme is independent of the active power level of the equipment and does not involve PLL. Furthermore, it does not alter the self-synchronization characteristics, inertial response characteristics, and SC energy distribution characteristics of the energy synchronization link. The MCLC technique can limit the current magnitude synthesized from the reactive and active currents and their respective magnitudes to

ensure the safety of the MMC and SCs. Theoretical analysis, comparative simulations, and experimental results obtained on a practical 35kV/30MVar/3MW/4s ET-STATCOM prototype have indicated the promising performance of the techniques. In addition, the transient stability analysis and stabilization mechanisms of the proposed strategy represent key directions for future research to further validate and enhance this study.

APPENDIX A

THE EXPRESSION OF THE MATRIX A

Unnumbered equation shown at the top of this page.

APPENDIX B

DETAIL CIRCUIT PARAMETERS OF THE STUDIED 35kV/30MVAR/3MW/4s ET-STATCOM

TABLE I
CIRCUIT PARAMETERS

	Parameters	Values
System parameters	Rated active power P_0	3 MW
	Rated reactive power Q_0	30 MVar
	Grid line-to-line voltage u_g	3 kV
	Rated DC bus voltage U_{dc0}	3 kV
	Grid frequency f_g	50 Hz
	Number of SMs per phase N	45
	FC Capacitance C_f	10 mF
	Arm Inductance L_s	30 mH
SC parameters	SCR	2
	Rated current of SC within 4 s	0.75 kA
	Maximum current of SC cell	1.5 kA
	Rated voltage of SC cell	2 V
	Maximum voltage of SC cell	2.8 V
	Number of SC cells per module	48
	Number of cascaded SC modules	40
	Rated capacitance per SC module	63 F
	Internal resistance of per SC cell	12 mΩ

REFERENCES

- [1] B. Kroposki et al., "Achieving a 100% renewable grid: Operating electric power systems with extremely high levels of variable renewable energy," *IEEE Power Energy Mag.*, vol. 15, no. 2, pp. 61–73, Mar./Apr. 2017.
- [2] H. Xiao, H. He, L. Zhang, and T. Liu, "Adaptive grid-synchronization based grid-forming control for voltage source converters," *IEEE Trans. Power Syst.*, vol. 39, no. 2, pp. 4763–4766, Mar. 2024.
- [3] Y. Zhou, D. Zhu, J. Hu, X. Zou, and Y. Kang, "Phase step control in PLL of DFIG-based wind turbines for ultrafast frequency support," *IEEE Trans. Power Electron.*, vol. 40, no. 1, pp. 51–56, Jan. 2025.
- [4] J. Rocabert, A. Luna, F. Blaabjerg, and P. Rodríguez, "Control of power converters in AC microgrids," *IEEE Trans. Power Electron.*, vol. 40, no. 1, pp. 4734–4749, Nov. 2012.
- [5] H. Liao et al., "Comprehensive-contribution-based primary frequency regulation market design for the converter-integrated power system," *IEEE Trans. Power Syst.*, vol. 39, no. 2, pp. 3825–3838, Mar. 2024.
- [6] G. Modi and B. Singh, "A control to address issues of synchronization and voltage rise in grid integrated solar photovoltaic-battery storage system," *IEEE Trans. Power Electron.*, vol. 40, no. 1, pp. 2385–2396, Jan. 2025.
- [7] S. Mohanty, A. Das, and B. Singh, "Virtual inertia control for CHB-STATCOM in a power electronics dominated grid," *IEEE Trans. Ind. Appl.*, vol. 59, no. 5, pp. 5517–5526, Sep./Oct. 2023.
- [8] X. Jang et al., "Low-frequency oscillation damping of grid-forming STATCOMs with phase-compensated DC-link voltage synchronization," *IEEE Trans. Power Electron.*, vol. 40, no. 2, pp. 2860–2873, Feb. 2025.
- [9] J. Fang, H. Li, Y. Tang, and F. Blaabjerg, "Distributed power system virtual inertia implemented by grid-connected power converters," *IEEE Trans. Power Electron.*, vol. 33, no. 10, pp. 8488–8499, Oct. 2018.
- [10] M. Zhang, M. Li, and J. Zhang, "Analytic model predictive current control of grid-connected power conversion system in battery energy storage system," *IEEE Trans. Power Electron.*, vol. 40, no. 3, pp. 3952–3967, Mar. 2025.
- [11] J. Wang et al., "Precise equivalent circuit model for Li-ion battery by experimental improvement and parameter optimization," *J. Energy Storage*, vol. 52, 2022, Art. no. 104980.
- [12] J. Liu, Y. Hou, J. Guo, X. Liu, and J. Liu, "A cost-efficient virtual synchronous generator system based on coordinated photovoltaic and supercapacitor," *IEEE Trans. Power Electron.*, vol. 38, no. 12, pp. 16219–16229, Dec. 2023.
- [13] X. Shi, C. Liu, C. Zhang, R. Li, and X. Cai, "Low-frequency current ripple minimization of single-star bridge cells-based battery energy storage system using optimal third harmonic voltage injection," *IEEE Trans. Energy Convers.*, vol. 38, no. 4, pp. 2525–2538, Dec. 2023.
- [14] Siemens Energy, "SVC PLUS FS (E-STATCOM)," 2022. [Online]. Available: <https://www.siemens-energy.com/global/en/home/products-services/product/svcplus-frequency-stabilizer.html>
- [15] Hitachi Energy, "SVC light enhanced," 2021. [Online]. Available: <https://www.hitachienergy.com/products-and-solutions/facts/statcom/svc-light-enhanced>
- [16] Sieyuan, "Static synchronous condenser (E-STATCOM)," 2024. [Online]. Available: https://mp.weixin.qq.com/s/L8b2gejfZu9jLlx_ZcJH_g
- [17] NR Electric, "Static synchronous condenser (SSC)," 2024. [Online]. Available: <https://mp.weixin.qq.com/s/Qlfcx8gy81gZ8X4Lr4ShLA>
- [18] Y. Wu, H. Wu, F. Zhao, Z. Li, and X. Wang, "Influence of PLL on the stability of interconnected grid-forming and grid-following converters," *IEEE Trans. Power Electron.*, vol. 39, no. 10, pp. 11980–11985, Oct. 2024.
- [19] X. Wu et al., "Grid-forming control and experimental validation for high voltage transformerless battery energy storage system," *IEEE Trans. Power Electron.*, vol. 40, no. 4, pp. 4889–4901, Apr. 2025.
- [20] P. Sun, Z. Tian, M. Huang, X. Zha, X. Li, and W. Wang, "Additional kinetic energy injection and piecewise damping based postfault antiwindup and transient stability enhanced control for grid-forming inverter," *IEEE Trans. Power Electron.*, vol. 39, no. 7, pp. 8007–8023, Jul. 2024.
- [21] Q.-C. Zhong and G. Weiss, "Synchronverters: Inverters that mimic synchronous generators," *IEEE Trans. Ind. Electron.*, vol. 58, no. 4, pp. 1259–1267, Apr. 2011.
- [22] L. Zhang, L. Harnefors, and H.-P. Nee, "Power-synchronization control of grid-connected voltage-source converters," *IEEE Trans. Power Syst.*, vol. 25, no. 2, pp. 809–820, May 2010.
- [23] R. Aouini, B. Marinescu, K. B. Kilani, and M. Elleuch, "Synchronverter based emulation and control of HVDC transmission," *IEEE Trans. Power Syst.*, vol. 31, no. 1, pp. 278–286, Jan. 2016.
- [24] Z. Zhao, P. Yang, Y. Wang, Z. Xu, and J. M. Guerrero, "Dynamic characteristics analysis and stabilization of PV-based multiple microgrid clusters," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 805–818, Jan. 2019.
- [25] S. Sang, C. Zhang, X. Cai, M. Molinas, J. Zhang, and F. Rao, "Control of a type-IV wind turbine with the capability of robust grid-synchronization and inertial response for weak grid stable operation," *IEEE Access*, vol. 7, pp. 58553–58569, 2019.
- [26] L. Zhao, Z. Jin, and X. Wang, "Analysis and damping of low-frequency oscillation for DC-link voltage-synchronized VSCs," *IEEE Trans. Power Electron.*, vol. 38, no. 7, pp. 8177–8189, Jul. 2023.
- [27] C. Arghir and F. Dörfler, "The electronic realization of synchronous machines: Model matching, angle tracking, and energy shaping techniques," *IEEE Trans. Power Electron.*, vol. 35, no. 4, pp. 4398–4410, Apr. 2020.
- [28] L. Huang et al., "A virtual synchronous control for voltage-source converters utilizing dynamics of DC-link capacitor to realize self-synchronization," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 5, no. 4, pp. 1565–1577, Dec. 2017.
- [29] Y. Qin, H. Wang, Z. Deng, J. Zhang, R. Yang, and X. Cai, "Control of inertia-synchronization controlled wind turbine generators under symmetrical grid faults," *IEEE Trans. Energy Convers.*, vol. 38, no. 2, pp. 1085–1096, Jun. 2023.
- [30] D. U. Sarkar, T. Prakash, and S. N. Singh, "Fractional order PID-PSS design using hybrid deep learning approach for damping power system oscillations," *IEEE Trans. Power Syst.*, vol. 40, no. 1, pp. 543–555, Jan. 2025.
- [31] N. Baekeland, C. Debjyoti, M. Lu, B. Johnson, and G. S. Seo, "Overcurrent limiting in grid-forming inverters: A comprehensive review and discussion," *IEEE Trans. Power Electron.*, vol. 39, no. 11, pp. 14493–14517, Nov. 2024.
- [32] X. Shi, R. Yang, X. Cai, Z. Fang, P. Dong, and F. Rao, "Improved comprehensive energy-based control for MMC-HVDC system," *Int. J. Elect. Power Energy Syst.*, vol. 145, 2023, Art. no. 108593.
- [33] S. Milovanović and D. Đujić, "Comprehensive comparison of modular multilevel converter internal energy balancing methods," *IEEE Trans. Power Electron.*, vol. 36, no. 8, pp. 8962–8977, Aug. 2021.
- [34] M. G. Taul, X. Wang, P. Davari, and F. Blaabjerg, "Current limitation control with enhanced dynamics of grid-forming converters during fault conditions," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 8, no. 2, pp. 1062–1073, Jun. 2020.
- [35] Y. Zhang, C. Zhang, M. Molinas, R. Yang, and X. Cai, "Current-constrained power-angle characterization method for transient stability analysis of grid-forming voltage source converters," *IEEE Trans. Energy Convers.*, vol. 38, no. 2, pp. 1338–1349, Jun. 2023.
- [36] Y. Tao, Q. Liu, Y. Deng, X. Liu, and X. He, "Analysis and mitigation of inverter output impedance impacts for distributed energy resource interface," *IEEE Trans. Power Electron.*, vol. 30, no. 7, pp. 3563–3576, Jul. 2015.
- [37] J. Wang and K. Ma, "Inertia and grid impedance emulation of power grid for stability test of grid-forming converter," *IEEE Trans. Power Electron.*, vol. 38, no. 22, pp. 2469–2480, Feb. 2023.
- [38] F. Wang, J. Xu, and G. Li, "A variable virtual impedance current limitation strategy of grid-forming energy storage-STATCOM," *IEEE Trans. Power Del.*, vol. 39, no. 6, pp. 3450–3461, Dec. 2024.
- [39] B. Fox et al., *Wind Power Integration: Connection and System Operational Aspects*, Stevenage, U.K.: Inst. Eng. Technol., 2008.
- [40] *Requirements for micro-generating plants to be connected in parallel with public low-voltage distribution networks*, NEN-EN 50438, 2013.
- [41] L. Harnefors, M. Hinkkanen, U. Riaz, F. M. M. Rahman, and L. Zhang, "Robust analytic design of power-synchronization control," *IEEE Trans. Ind. Electron.*, vol. 66, no. 8, pp. 5810–5819, Aug. 2019.
- [42] Q. Qiu, X. Xiang, J. Lei, W. Li, and X. He, "Transient stability analysis for paralleled system of virtual synchronous generators based on damping energy visualization and approximation," *IEEE Trans. Power Electron.*, vol. 39, no. 12, pp. 15785–15799, Dec. 2024.
- [43] X. Lyu, W. Du, S. M. Mohiuddin, and Y. Cheng, "Critical clearing time for droop-controlled grid-forming inverters with circular current limiting and virtual impedance current limiting," *IEEE Trans. Power Syst.*, vol. 40, no. 2, pp. 1997–2000, Mar. 2025.